

Java

Methods in Java :-

2. Non-Static Method :-

- A non-static method belongs to an object of the class.
- It can access both static and non-static members of the class.
- To call it, you need to create an object of the class.
- It is also called an instance method.

Syntax –

```
class ClassName {  
    // non-static method  
    returnType methodName(parameters) {  
        // method body  
    }  
}
```

Example –

```
package Basic_Concept;  
  
public class nonstatic_method {  
    public static void main(String[] args) {  
        //1.creating object  
        nonstatic_method obj=new nonstatic_method();  
  
        //2.call nonstatic_method using object  
        obj.add();  
    }  
  
    void add()  
    {  
        System.out.println("Addition is: "+(4+8));  
    }  
}
```

Output –

Addition is: 12

3. Parameterized Method :-

- A method that accepts parameters (inputs) when called.
- Parameters allow the method to receive values from outside.
- Makes the method more flexible and reusable.
- You can call the same method with different arguments.

Syntax –

```
returnType methodName(parameterType parameter1, parameterType parameter2, ...) {  
    // method body  
}
```

Example –

```
package Basic_Concept;  
  
public class parameterized_method {  
    public static void main(String[] args) {  
        //call static method  
        subtraction(40,20);  
  
        //call non-static method  
        parameterized_method obj=new parameterized_method();  
        obj.multiplication(3,4);  
    }  
  
    //define static method  
    static void subtraction(int n1,int n2)  
    {  
        System.out.println("Substraction is: "+(n1-n2));  
    }  
  
    //define non-static method  
    void multiplication(int a,int b)  
    {  
        System.out.println("Multiplication is: "+(a*b));  
    }  
}
```

Output –

Substraction is: 20

Multiplication is: 12

4. Return Type :-

- A method that returns a value after execution.
- The return type specifies the type of value the method will give back (e.g., int, String, double).
- If a method does not return anything, its return type is void.
- The return keyword is used to send the value back to the caller.

Syntax –

```
returnType methodName(parameters) {  
    // method body  
    return value; // returning value of returnType  
}
```

Example –

```
package Basic_Concept;  
  
public class return_type {  
    public static void main(String[] args) {  
        //call static method  
        int num=add(10,20);  
        System.out.println("Addition is: "+num);  
  
        //call non-static method  
        return_type obj=new return_type();  
        String msg=obj.str("Sagar");  
        System.out.println(msg);  
    }  
  
    //define static method  
    static int add(int a,int b)  
    {  
        return a+b;  
    }  
  
    //define non-static method  
    String str(String name)  
    {  
        return name;  
    }  
}
```

Output –

Addition is: 30

Sagar